

STUDYING THE SUN FROM SPACE

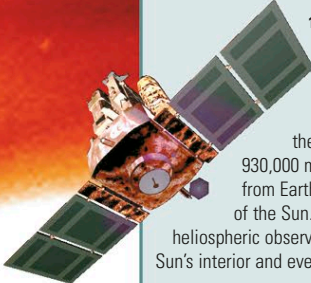
Since 1960, a series of space probes and satellites have been launched by NASA and other organizations with the aim of collecting data about the Sun. Some of the most important missions are listed below.

1960–1968 PIONEERS 5 TO 9 (USA) These were a series of probes that successfully orbited the Sun and studied the solar wind, solar flares, and the interplanetary magnetic field.

1974, 1976 HELIOS 1 AND 2 (USA AND GERMANY) The two Helios probes were put into orbits that involved high-velocity passes close to the Sun's surface. They measured the solar wind and the Sun's magnetic field.

1980 SOLAR MAXIMUM MISSION (USA) This studied the Sun at its most active, collecting X-rays, gamma rays, and ultraviolet radiation produced by flares, sunspots, and prominences.

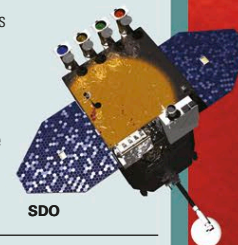
1990 ULYSSES (USA AND EUROPE) The first space probe to be sent into an orbit over the Sun's poles, Ulysses has studied the solar wind and the Sun's magnetic field over its polar regions.



1995 SOHO (USA AND EUROPE) This solar observatory follows a special "halo" orbit around the Lagrangian point 930,000 miles (1.5 million km) from Earth in the direction of the Sun. SOHO (solar and heliospheric observatory) studies the Sun's interior and events at its surface.

2006 STEREO (USA) The Solar Terrestrial Relations Observatory consists of twin spacecraft that observe the Sun from different directions, giving all-around coverage of solar eruptions and the solar wind.

2010 SDO (USA) NASA's Solar Dynamics Observatory monitors the Sun to improve our understanding of its activity and to make better predictions of how this activity will affect Earth.



2018 PARKER SOLAR PROBE (USA) Using Venus flybys to slow itself down, Parker will orbit the Sun as close as 3.8 million miles (6.1 million km) to study the Sun's atmosphere.

2020 SoI0 (EUROPE AND USA) From an orbit just inside Earth's orbit to just inside Mercury's orbit, the Solar Orbiter will study the solar wind and the Sun's poles.

